Fiscal Year:	FY 2004	**Task Last Updated:**	FY 04/03/2006
PI Name:	Reschke, Millard F Ph.D.		
Project Title:	A Countermeasure for Space Motion Sickness		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058-3607	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	06/01/2004	**End Date:**	06/01/2007
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	start and end dates are uncertain		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
	Problems associated with space motion sickness (SMS) and clear vision (VOR changes - retinal slip) during space flight have no realistic solution given existing countermeasures (including: medications for symptoms associated with microgravity, preflight adaptation, and planned on orbit centrifugation). For example, drugs can only be used at specific times during a flight, they cannot prevent the appearance of motion sickness symptoms, and regardless of their efficacy, all drugs have undesirable side effects. Preflight adaptation, while it appears to be effective, requires substantial investments in crew time, and crewmembers must experience some level of motion sickness if the preflight adaptation environment is a serious analog of flight. Inflight centrifugation may be the ultimate vestibular countermeasure, however, rapid implementation is impracticable, and like preflight adaptation protocols, crewmembers will experience motion sickness on both ground and flight centrifuges. The purpose of this proposal is to investigate both the prevention		

Task Description:	of motion sickness and retinal slip through the use of stroboscopic vision. The research for this proposal is divided into four phases. The Phase I protocol requires a three-part research plan that will: (1) verify the effectiveness of stroboscopic vision in preventing motion sickness induced with left-right visual reversing prisms; (2) investigate the effect of flash rate and duration on prism induced motion sickness; and (3) study the adaptation of motion sickness susceptibility under stroboscopic illumination. In Phase II, stroboscopic goggles using LCD electronic shutters will be constructed to replace the stroboscopic flash generator for use in environments where ordinary stroboscopic flash is not feasible. In Phase III the effect of these goggles will be verified; then tested in unique provocative environments; including both parabolic flight and tests within the Navy's Slow Rotation Room centrifuge. Finally, in Phase IV the probable mechanisms involved with stroboscopic presentation and the prevention of motion sickness and retinal slip will be investigated using the newly constructed stroboscopic goggles. These investigations are directly responsive to the goals of NSBRI's neurovestibular research program: To develop countermeasures that help crewmembers: avoid disorientation, meet the physical requirements for emergencies, treat motion sickness without side effects, and assure safe control of the spacecraft.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	New project for FY2004.
Bibliography Type:	Description: (Last Updated: 11/14/2025)